Addressing Data Scarcity in Financial Loan Contract NER Using Synthetic LLM-Generated Data

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Abstract

Named Entity Recognition (NER) in financial loan contracts facilitates the extraction of relevant contractual information that can later be used in financial analysis. To extract information from loan contracts, transformer-based language models can be employed; however, their performance in financial domains often depends on domain-specific fine-tuning, since confidential loan agreements are underrepresented in publicly available pretraining corpora.

In this work, we investigate the effectiveness of fine-tuning transformer-based models on synthetic datasets generated and annotated using Large Langugae Models (LLMs). Although this work focuses on financial loan agreements between corporations, the findings may also provide insights into the use of LLM-generated data for NER tasks across other legal document domains.

1 Introduction

Financial risk assessment plays a critical role in the stability of global economies. The absence of reliable risk models can contribute to widespread financial instability, as was evident in the 2006-2008 financial crisis. To support risk assessment, financial institutions increasingly use Artificial Intelligence (AI) driven tools to analyze unstructured data (Crouhy, 2010; Banking Vision, 2025).

Accurate recognition of key entities enables the association of contractual terms with the relevant parties, thereby supporting downstream risk assessment. However, information extraction models typically require large amounts of annotated data to achieve reliable performance (Zin et al., 2023), which is often unavailable for loan agreements due to their confidential nature. Additionally, data annotation is time consuming, requires domain experts, and is expensive (Nadăș et al., 2025).

Pretrained LLMs are exposed to broad and diverse corpora (Devlin et al., 2019) that includes legal, financial, and regulatory text. Through this exposure, the modern models learn generalizable representations of legal language, document structure, and financial terminology. As a result, they are capable of generating synthetic loan agreements that approximate key characteristics of real contracts (Nadăș et al., 2025). In our work, we investigate whether LLM generated and annotated domain specific datasets can support the limited public availability of annotated loan agreements. This leads us to the following research question:

How effective are synthetic annotated datasets for data augmentation in information extraction tasks involving financial loan agreements?

2 Related Work

Early work on domain adaptation (Alvarado et al., 2015) showed that combining large-scale labeled newswire data (CoNLL-2003[1]) with a publicly available set of financial agreements from the U.S. Securities and Exchange Commission (SEC) improves NER model training. The authors also observed that their best-performing model struggled to distinguish between organization (recall: 0.513) and location (recall: 0.286) entities due to their contextual similarity. Since BERT, introduced after this work (Devlin et al., 2019), provides stronger contextual representations and better captures long-range dependencies than the traditional feature-based methods used in the study, we investigate how transformer-based model performs under the same target-domain dataset setting.

Additionally, recent studies have shown that synthetic data can mitigate data scarcity and reduce annotation costs, while also providing greater control over dataset composition and improved privacy preservation (Nadăș et al., 2025; Li et al., 2023). Other work has explored methods for efficiently

[1] https://huggingface.co/datasets/eriktks/conll2003

generating synthetic training data using large language models (Li et al., 2026). In the context of financial NLP, these findings motivate investigating whether data scarcity can be addressed by generating synthetic loan agreements for data augmentation. Prior work has also highlighted the potential of LLM-based annotation to reduce manual labeling effort while maintaining competitive performance, which motivates further analysis of how modern approaches affect model performance in this setting.

3 Methodology

3.1 Model

To evaluate the effect of synthetic data, we use the general-purpose BERT model `google-bert/bert-base-cased`[2] rather than domain-specific pretrained alternatives. This choice helps ensure that performance differences can be attributed to the fine-tuning data, as domain-specific models may introduce confounding effects from prior domain knowledge.

3.2 Datasets

3.2.1 In-domain financial loan agreement dataset (FIN5/FIN3)

The target-domain datasets (FIN5/FIN3) contain 54,256 tokenized words and consist of eight debt agreements between companies and financial institutions. The dataset was tokenized in a previous study (Alvarado et al., 2015) and manually annotated in CoNLL format with four named entity types: location (LOC), organization (ORG), person (PER), and miscellaneous (MISC), along with part-of-speech (POS) tags. The entity annotations follow an IO tagging scheme using only the I- prefix. Additionally, all occurrences of the tokens `Lender` and `Borrower` were labeled as PER. Although these entities typically refer to organizations, the PER label was retained as a generic category for agreement participants to preserve consistency with prior work.

The eight financial agreements are split into two subsets: FIN5 (five documents for tuning) and FIN3 (three documents used for testing). The documents vary in length, ranging from approximately 3,000 to 18,000 words (see Appendix A Figure 1a). Entity distributions are also highly imbalanced, with most contracts containing more PER entities than LOC or ORG, and varying distributions across documents (see Appendix A Figure 1b). This heterogeneity reflects the inherently contract-specific nature of loan agreements, which are shaped by individual negotiation terms and credit conditions between parties.

For model development, the FIN5 dataset was later split into a training set containing four contracts and a validation set containing one contract.

3.2.2 Out-of-domain control dataset (EWT)

To establish an out-of-domain baseline, we used the Universal NER dataset from the Universal NER project[3], a large CoNLL-formatted corpus with BIO tagging for PER, LOC, and ORG entities. We use it as a sanity check to verify that our training and evaluation pipeline is correctly implemented, while also evaluating the control model on the FIN3 dataset.

3.3 Datasets preprocessing

The in-domain dataset (FIN5/FIN3) originally uses an IO tagging scheme and includes POS tags designed for feature-engineered models; however, since transformer-based models learn contextual information directly from raw text, POS tags were removed. We further convert the labels to a BIO scheme for clearer entity boundaries and standard NER compatibility. Lastly, to ensure consistency and comparability with the control dataset, we map the rare MISC entities (10 tokens in total) to O, focusing the evaluation on the more relevant PER, LOC, and ORG entities.

3.4 Synthetic Dataset Generation Pipeline

3.4.1 LLM Selection

We used instruction-tuned models and manually evaluated outputs based on formality, repetition, entity diversity, and coherence to select the best-performing model for contract generation. Smaller models (e.g., Gemma-2B-IT) struggled with multi-constraint prompts, so we tested larger baselines and selected Mistral-7B-Instruct-v0.3[4] for its stronger instruction adherence, further improved through iterative prompt engineering (Appendix A, Figure 4).

[2] https://huggingface.co/google-bert/bert-base-cased

[3] https://www.universalner.org/

[4] https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3

3.4.2 Synthetic Contract Generation

Generation set up

Synthetic loan agreements were generated using a structured multi-prompt engineering approach (see Appendix A Figure 3), following recent studies (Long et al., 2024). The strategy included explicit LLM role specification (e.g., "You are a legal document planner") and format instructions, mandatory section requirements, and conditional constraints such as language style and number of sections.

Generating complete contracts in a single step is challenging, as long output generation is known to suffer from instruction drift (Li et al., 2026). For this reason, full contract generation was decomposed into planning and section-level synthesis stages (see Appendix B for the planning and section generation prompts).

In the first stage, the model generated a structured contract plan in JSON format, containing the contract description, metadata, and a list of sections with brief summaries. Outputs were validated for correct JSON structure and required fields to ensure usability in downstream tasks. The process allowed up to three attempts before termination upon repeated failure. In practice, most errors were resolved on the second attempt, while complete failure within three attempts was rare (2 out of 11 cases).

Once a valid plan was obtained, the contract was generated section by section using a section-prompt template, where each step included the first stage generated global plan, text from previous sections for continuity, and the objective of the current section.

Prompts polishing and adjustments

Prompts were iteratively refined based on manual inspection of the generated outputs. As a result, the final prompts include multiple constraints and requirements that emerged through this iterative improvement process.

Temperature was set to 0.8 for planning to promote entity diversity and 0.6 for contract generation to maintain more deterministic legal text style. Despite this, while overall wording diversity remained acceptable, entity diversity was limited, with repeated placeholders such as `ABC Company` and overrepresented locations such as `New York` and `California`, increasing sample similarity and raising concerns about overfitting in downstream training.

To mitigate this issue, we introduced the `Faker()`[5] library, which contains large internal datasets of entities and generates data via random sampling. In the planning stage, attributes such as locations, contact information, and dates were generated using `Faker()` and provided to the LLM, while organization names were excluded as they were often overly simplistic. This contextual information led the model to infer more realistic and diverse organization names.

Final contracts were similar in length and entity distribution (see Appendix A Figure 2a and Figure 2b). Compared to FIN5, entity proportions remained broadly consistent, with PER appearing more frequently than LOC and ORG across documents. However, synthetic contracts were not only shorter than FIN5 documents, but also more homogeneous in both size and entity distribution, making them less diverse between each other.

Limitations

While the generated contracts were sufficient for our purposes and followed a realistic loan-agreement structure with the required entities, they were not fully free of imperfections. The main issues included unrealistic locations, formatting artifacts such as echo prefixes (`Output:`, `Section:`, `Result:`), and repetitive sentences due to limited context between independently generated sections. These limitations are common in synthetic data generation but still provide useful training signal despite reduced surface-level realism. To address them, we reviewed the synthetic contracts and removed present echo noise.

3.4.3 Model Selection for Annotation

To generate annotations, we evaluated both `Mistral 7B` and `GPT-5.4` [6] as candidate annotators. To compare the models, we used a single contract sample from the FIN5 dataset for annotation and validation. We used one-shot prompting by providing an example input-output pair (see example pair in Appendix B Figure 9).

`GPT-5.4` outperformed `Mistral 7B` across all metrics (Appendix A, Table 5). Manual review confirmed that Mistral often failed to follow the required format, causing token-label misalignment, whereas `GPT-5.4` consistently produced well-structured outputs, leading to its selection as

[5] https://faker.readthedocs.io/en/master/

[6] https://openai.com/index/introducing-gpt-5-4/

the synthetic data annotator.

The model was accessed through authenticated API calls, allowing the annotation pipeline to be automated efficiently. Despite using a high-performance commercial model, the total cost of annotation experiments and synthetic data generation remained relatively low, at approximately $5 for $\approx$ 1M tokens in API usage.

3.4.4 Data Annotating

As mentioned before, to produce LLM-annotations, we used the one-shot prompting technique, which in this case corresponds to a selected sentence from the FIN5 dataset. Additional instructions in the prompt mostly indicate formatting constraints, and establish the `B-PER` tagging convention for the entities `Lender` and `Borrower` (see Appendix B Listing 3).

Prior to annotation, data was tokenized and prepared to display the output format `token<TAB>label<\n>`. This format was selected for several reasons: (1) It is the format used in the annotated datasets. (2) The LLM writes each token directly next to its annotation, which improves alignment reliability. (3) It allows easier verification of which annotation belongs to each token.(4) More complex formats such as nested list structures (e.g., `["token", "label"]`) were also tested, but resulted in excessively many malformed outputs, including unmatched brackets and skipped tokens.

To improve the efficiency of API usage and ensure the quality of the generated annotations, two additional strategies were applied:

1. Multiple sentences were grouped into a single API call to reduce costs. This was achieved by tracking a cumulative token count using a parameter called `max_chunk_size`, set to 50. As grouping removed sentence boundaries, metadata for these "super-sentences" (referred to as *chunks*) had to be maintained to reconstruct the original sentence structure.

2. Imperfect outputs from the LLM were accepted but required validation to ensure no tokens were skipped. Missing tokens were labeled as `MISMATCHING`, allowing identification and manual revision during later data curation steps.

3.5 Comprehensive Performance Evaluation

3.5.1 Baseline

To establish the baseline, we fine-tuned `google-bert/bert-base-cased` on the control dataset `en_ewt-ud-train` using standard BERT model. The full training configuration is summarized in Table 1. The model was trained and evaluated on `en_ewt-ud` datasets to verify implementation correctness and establish baseline performance. It achieved a balanced precision and recall with an approximate Span F1 score of 0.69, providing a solid reference point for subsequent experiments.

Hyperparameter	**Value**
Model	bert-base-cased
Dataset	en_ewt-ud-train
Max sequence length	128
Learning rate	2e-5
Batch size	15
Epochs	8
Optimizer	AdamW

Table 1: Baseline model training configuration.

3.5.2 In-Domain model tuning

We fine-tuned the BERT model on in-domain FIN5 training data and further optimized its performance by adjusting the `max_length` parameter. This modification was necessary because the baseline model was trained on generic datasets with shorter sentences, whereas loan contracts typically contain longer sentences (see Appendix A, Figure 6). The best results were obtained with `max_length` set to 400, enabling the model to effectively capture longer contextual dependencies. To address potential overfitting, we also experimented with dropout regularization; however, it did not yield a substantial performance improvement and was therefore not included in the final setup (see tuning results in Appendix A, Table 6).

3.6 Data Augmentation Impact

We evaluated different levels of synthetic augmentation by progressively adding synthetic contracts to FIN5 and testing on FIN3. Span-F1 improved from 0.76 (FIN5 only, $\sim$1200 sentences) to 0.81 after adding six synthetic contracts ($\sim$2000 sentences), but decreased to 0.78 after adding the remaining three contracts (Appendix A, Figure 7). This together with almost perfect span performance on the training set suggests overfitting with more synthetic data. Therefore, the remaining analysis uses

Actual	Predicted				Recall
	PER	**LOC**	**ORG**	**NaNE**	
PER	165	0	0	5	0.971
LOC	0	19	0	16	0.543
ORG	0	7	21	20	0.438
NaNE	14	8	34	–	–
Precision	0.922	0.559	0.382	–	

Table 2: Entity-level confusion matrix for the model trained on FIN5 and evaluated on FIN3 under exact-match evaluation. ("NaNE" = Not a Named Entity)

Actual	Predicted				Recall
	PER	**LOC**	**ORG**	**NaNE**	
PER	162	0	0	8	0.953
LOC	0	22	0	13	0.629
ORG	2	0	20	26	0.417
NaNE	4	13	14	–	–
Precision	0.964	0.629	0.588	–	

Table 3: Entity-level confusion matrix for the model trained on a combination of FIN5 and synthetic data, evaluated on FIN3 under exact-match evaluation.

FIN5 + six synthetic contracts, which achieved the best performance. To further analyze the generated data, we measured entity diversity for each entity type as:

$$D(e) = \frac{N_{unique}}{N_{total}}$$

and compared the values across FIN5, synthetic and mixed datasets (Appendix A, Figure 8). The synthetic dataset shows lower PER and LOC diversity than FIN5, although ORG diversity is higher. In the mixed dataset, ORG diversity further increases (0.210 to 0.338), while PER diversity drops substantially (0.033 to 0.007).

4 Results and Discussion

The results were analyzed using both span and entity-level metrics under exact and loose matching criteria, where loose matching allows partial span overlap with the correct entity type. Table 4 reports only exact-match span-level results for conciseness.

Tune	Test	F1	Recall	Precision
Control	Control	0.69	0.68	0.71
Control	FIN3	0.17	0.13	0.28
FIN5	FIN3	0.76	0.81	0.71
Synthetic	FIN3	0.68	0.67	0.69
FIN5+Syn	FIN3	0.81	0.81	0.81

Table 4: Overall span-level exact-match results.

Preliminary results show that domain-specific fine-tuning greatly improves performance, with the FIN5 model achieving an exact-match F1 of 0.76 compared to 0.17 for the baseline model. This difference is mainly driven by PER entities, where recall increases from 0.035 to 0.971 (see Table 2 and Appendix A Table 7), likely due to consistent Borrower and Lender annotations across FIN5 and FIN3. The small difference between strict and loose PER metrics suggests that these entities are usually short and easy to localize. In contrast, both models show weak recall for LOC and ORG entities, while improved loose match scores indicate difficulties in exact span boundary detection for longer entities.

Training on synthetic data only reduces performance compared to training on FIN5, with a particularly large drop in ORG recall, decreasing from 0.438 (Table 2) to 0.104 (Appendix A, Table 8), which is even lower than the baseline recall of 0.208. Performance also decreases for PER and LOC entities, lowering the overall exact-match span F1 from 0.76 to 0.68.

Finally, combining real in-domain data with synthetic data produced the best overall results. Although PER and ORG recall slightly decreased compared to training on FIN5 only, LOC benefited substantially, with strict recall increasing from 0.543 (Table 2) to 0.629 (Table 3) and loose recall reaching 0.829. As a result, exact-match span F1 improved from 0.76 to 0.81 (Table 4). Compared to the values reported by (Alvarado et al., 2015), LOC recall increases from 0.286 to 0.629 and ORG recall decreases from 0.513 to 0.417. Further inspection of test set predictions (Appendix B, Listing 4) suggests that the model struggles with uppercase entity formats, while capitalized names are more reliably detected. The examples also reveal annotation inconsistencies in FIN3, where some correctly predicted entities are missing from the gold labels.

5 Conclusions

Results show that domain specific fine-tuning substantially improves performance over a baseline model trained on general domain data, particularly for PER entities. Combining FIN5 with synthetic contracts further improved overall performance, especially for LOC entities, although the gains remained relatively modest considering the amount of generated data. Our analysis suggests that entity diversity alone does not explain the results, since ORG entities show the highest diversity while also obtaining the weakest recall. This may instead be related to differences in entity complexity and formatting between datasets, such as shorter synthetic entities ("Acme Inc") versus longer FIN3 entities ("Silicium de Provence SAS"), as well as the limited presence of uppercase entities in the synthetic data. Inspection of prediction examples also revealed annotation inconsistencies across datasets, producing false positives that should likely be true positives.

Overall, these findings highlight the difficulty of generating high quality synthetic legal data and suggest that more advanced prompting or generation strategies could help produce more and realistic contracts, especially for longer and more complex entity spans, while maintaining annotation consistency.

Another important observation is that the high span scores achieved by GPT-5.4 on the FIN5 validation set raise the question of whether LLM-based extraction could itself replace more complex model training pipelines for information extraction tasks. However, when privacy or deployment constraints prevent the use of external APIs, fine-tuned open-weight models such as the BERT-based approach explored in this work remain a practical alternative.

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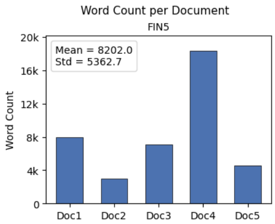
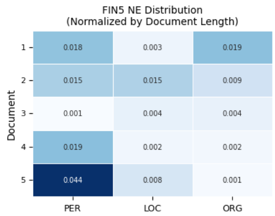
A Appendix: Figures and Results

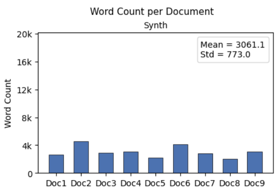


(a) Word count distribution

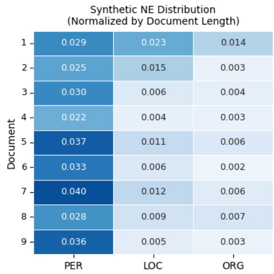


(b) Named entity distribution

Figure 1: Statistics of the FIN5 dataset showing (left) document length distribution and (right) named entity distribution.



(a) Word count distribution in synthetic data



(b) Named entity distribution in synthetic data

Figure 2: Statistics of the synthetic dataset showing (left) document length distribution and (right) named entity distribution.

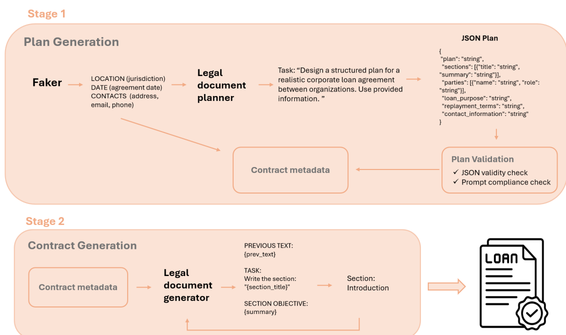


Figure 3: Overview of the two-stage synthetic contract generation pipeline. Stage 1 generates a structured plan using a legal document planner and validation module, producing JSON-formatted contract metadata. Stage 2 uses this metadata to guide a legal document generator that incrementally produces the final contract text.

```
<bos> Write an introduction of a Loan Agreement between two companies. The agreement
should include realistic names of the parties, loan amount, interest rate, repayment
schedule, and dates. The entire response must be a continuous block with no line breaks.

Gemma:

We are pleased to enter into this Loan Agreement (the "Agreement") between you,
ABC Company, and XYZ Company, for the sum of $10,000. The purpose of this Agreement is
to establish a clear understanding of the terms and conditions of the loan, including
the amount of the loan, interest rate, repayment schedule, and any other relevant
information. <...>

Mistral:

This Loan Agreement (the "Agreement") is entered into as of January 14, 2021, by and
between Northbridge Capital Partners Limited, a limited partnership organized under
the laws of the State of Delaware, with its principal place of business at 1234 Bank
Street, New York, NY 10001 (the "Lender"), and the following subsidiaries of ABC
Corporation, a Delaware corporation with its principal place of business at 5678
Capital Avenue, Chicago, IL 60601 (the "Borrowers"): <...>

Llama:

First National Bank of New York, a national banking association with its principal
place of business at 100 Wall Street, New York, NY 10005 (the "Lender"), and ABC
Corporation, a corporation with its principal place of business at 123 Main Street,
Anytown, USA 12345 (the "Borrower"), and XYZ Holding Company, a corporation with its
principal place of business at 456 Elm Street, Othertown, USA 67890 (the "Guarantor").
<...>

FIN5:

THIS LOAN AGREEMENT (" this Agreement ") is entered into by and between the parties
below in Beijing, China as of March 31, 2007: Thinkplus Investments Limited., a
corporation incorporated under the laws of the Cayman Islands, whose registered
address is Codan Trust Company (Cayman) Limited, Century Yard, Cricket Square,
Hutchins Drive, P. O. Box 2681GT, George Town, Grand Cayman, British West Indies,
hereinafter referred to as the "Company "; Airland International Limited, a
corporation incorporated under the laws of the British Virgin Islands, whose
registered address is 2nd floor, Abbott Building Road Town, Tortola, British
Virgin Islands, hereinafter referred to as the" Airland " <...>
```

Figure 4: Synthetic data generation example between LLMs.

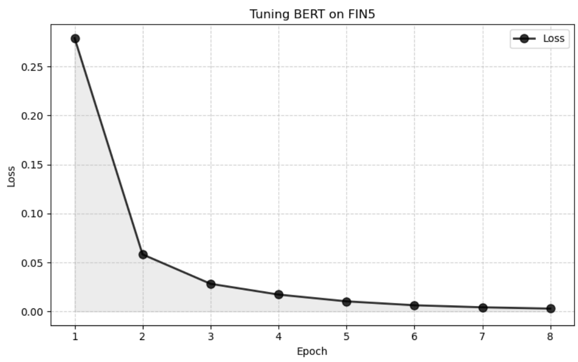


Figure 5: BERT tuning on financial dataset.

Metric	GPT-5.4	Mistral
Precision	0.91	0.06
Recall	0.95	0.07
F1	0.93	0.06
Accuracy	0.99	0.12

Table 5: Comparison of evaluation metrics between GPT-5.4 and Mistral-7B-Instruct-v0.3 performance on data annotation task.

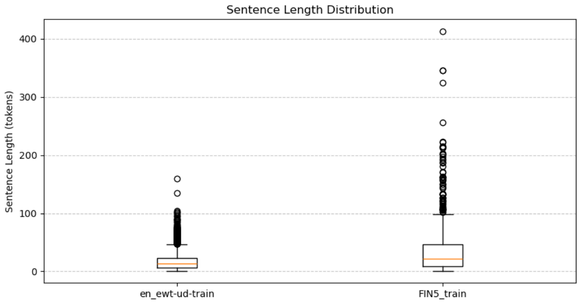


Figure 6: Sentence length comparison between control data and loan contracts.

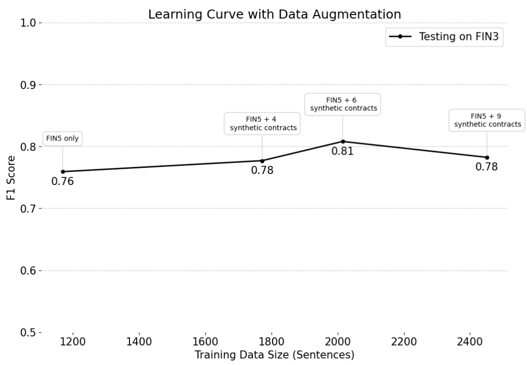


Figure 7: Learning curve showing the F1-Score as more synthetic data is added to the training dataset. Experiment starts in FIN5 and incrementally adding synthetic contracts.

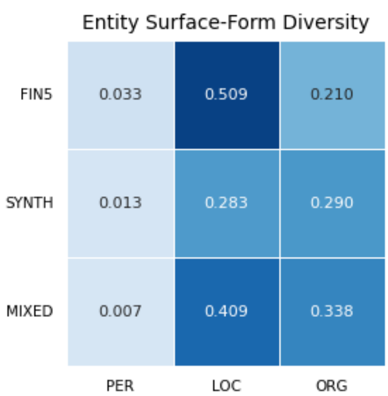


Figure 8: Entities diversity scores across different datasets. Higher scores representing datasets with more distinct entities.

Model	Recall	Precision	Span F1
128	0.84	0.77	0.81
256	0.87	0.83	0.85
400	0.92	0.88	0.90
400 + Dropout	0.93	0.87	0.90

Table 6: Model tuning for different maximum sequence lengths. A dropout rate of 0.2 performed best among the tested dropout settings and used for configuration testing. Model was tuned on FIN5 development set (4 contracts) and tested on FIN5 validation set (1 contract).

Actual	**PER**	**LOC**	**ORG**	**O**	**Recall**
PER	6	0	16	148	0.035
LOC	1	16	0	18	0.457
ORG	3	1	10	34	0.208
NaNE	3	5	24	–	–
Precision	0.462	0.727	0.200	–	

Table 7: Confusion matrix for entity-level evaluation. Model was trained on out-of-domain data and evaluated on FIN3.

Actual	**PER**	**LOC**	**ORG**	**O**	**Recall**
PER	150	0	0	20	0.882
LOC	0	15	0	20	0.429
ORG	2	6	5	35	0.104
NaNE	14	18	11	–	–
Precision	0.904	0.385	0.312	–	

Table 8: Confusion matrix for entity-level evaluation. Model was trained on synthetic data only and evaluated on FIN3.

B Appendix: Prompts and Outputs

Listing 1: Plan Generation Prompt

```
--- ROLE AND MAIN TASK

You are a legal document planner.

Design a structured plan for a realistic
     corporate loan agreement between
    organizations.

Use the following information to
    generate plan:

--- FAKER

Lender: generate fake organization name
    (Contact: {fake_contact_lender['
    address']}, {fake_contact_lender['
    email']}, {fake_contact_lender['
    phone']})
```

```
Borrower: generate fake company name (
    Contact: {fake_contact_borrower['
    address']}, {fake_contact_borrower['
    email']}, {fake_contact_borrower['
    phone']})
Jurisdiction: {jurisdiction}
Agreement Date: {agreement_date}
Interest Rate: {interest_rate}

--- PLAN GENERATION

Now generate a new plan in the exact
    same JSON format, using the
    information provided:

{
  "plan": "string",
  "sections": [{"title": "string", "
     summary": "string"}],
  "parties": [{"name": "string", "role":
      "string"}],
  "loan_purpose": "string",
  "replayment_terms": "string",
  "contact_information": "string"
}

Requirements:
- Corporate loan between a financial
    institution and one or more
    corporate borrowers
- Include multiple entities with clear
    roles (e.g., Lender, Borrower)
- Use realistic, fully specified company
     names

Sections:
- Must include: Introduction, Parties,
    Loan Terms, Definitions,
    Representations and Warranties,
    Interest and Fees, Termination,
    Signatures
- You may rename mandatory sections
- Add additional standard loan agreement
     sections
- Total sections: 10  15
- Each section must include a short
    summary

Plan:
- Describe parties, roles, loan purpose,
     jurisdiction, obligations, risks,
    and contact details
- Explicitly mention all parties by name

Constraints:
- Do NOT use placeholders (e.g., [
    Company], [Location], [X])
- Ensure consistency across sections
- Output JSON only
```

Listing 2: Section Generation Prompt

```
You are a legal document generator.

GLOBAL PLAN:
{global_plan}

PARTIES:
{parties}

JURISDICTION:
{jurisdiction}

AGREEMENT DATE:
{agreement_date}

LOAN PURPOSE:
{loan_purpose}

INTEREST RATE:
{interest_rate}

REPLAYMENT TERMS:
{replayment_terms}

CONTACT INFORMATION:
{contact_information}

ALL SECTIONS:
{sections_list}

PREVIOUS TEXT:
{prev_text}

TASK:
Write the section: "{section_title}"

SECTION OBJECTIVE:
{section_description}

GUIDELINES:
- Follow the GLOBAL PLAN closely
- Use the exact PARTY names and ROLES
    provided
- Maintain consistency with PREVIOUS
    TEXT
- Ensure terms (dates, amounts, roles,
    interest rate, loan purpose,
    replayment terms and so on) remain
    consistent with the context above

STYLE:
- Formal legal language
- Structured paragraphs
- Use subsections where appropriate

CONSTRAINTS:
- Do NOT repeat previous sections
- Do NOT include explanations
- Output ONLY the body text of this
    section
- Do NOT include the section title or
    numbering
```

Listing 3: Annotation Generation Prompt

```
You are a strict NER tagger.

Task:
Assign a BIO tag to EACH token.

Allowed labels:
B-PER, I-PER, B-ORG, I-ORG, B-LOC, I-LOC
    , O

Rules:
- Annotate legal party roles "Lender"
    and "Borrower" as B-PER
```

```
- EXACTLY one label per token
- SAME number of output lines as input
    tokens
- SAME token order as input
- Do not modify tokens
- Output format must be: token<TAB>label
- One token-label pair per line
- No explanations

Example input:
{example_input}

Example output:
{example_output}

Now annotate this input:
{{test_input}}
```

Listing 4: Comparison of labels between FIN3 and model predictions for a model trained on mixed data (six contracts). Outputs are shown as: token - FIN3 (gold) - model prediction.

```
--
    Token         FIN3    Model
3   Union         B-ORG   B-ORG
4   Bank          I-ORG   I-ORG
5   of            O       I-ORG
6   California    B-LOC   I-ORG
7   NA            O       I-ORG
8   and           O       O
9   Crocs         O       B-ORG
10  Inc           O       I-ORG
11  .             O       O
12  [             O       O

--
    Token        FIN3    Model
24  SILICIUM     B-ORG   O
25  DE           I-ORG   O
26  PROVENCE     I-ORG   O
27  SAS          I-ORG   O
28  and          O       O
29  EVERGREEN    B-ORG   O
30  SOLAR        I-ORG   O

    Token        FIN3    Model
4   Silicium     B-ORG   B-ORG
5   de           I-ORG   I-ORG
6   Provence     I-ORG   I-ORG
7   SAS          I-ORG   I-ORG
8   and          O       O
9   Evergreen    B-ORG   B-ORG
10  Solar        I-ORG   I-ORG
11  Inc          I-ORG   I-ORG
```

C Appendix: Contributions

Maribel Yazmin Preite: HPC setup, implementation of the GPT-5.4 and Mistral API pipelines for data annotation, LLM experimentation, synthetic data pipeline design, error analysis, data exploration, and model evaluation, report writing.

Gabija Makutenaite: HPC setup, model implementation, tuning, training, prediction generation and evaluation, data processing, synthetic data generation and optimization, report writing.

Ashanti Costa Martins: provided project support and contributed to discussions and team coordination throughout the project.

```
350
,
Wellesley
,
Massachusetts
02481
doing
business
as
"
Silicon
Valley
East
"
and
AKAMAI
TECHNOLOGIES
,
INC
(
"
Borrower
"),
```

Input prompt

```
350 O
, O
Wellesley B-LOC
, O
Massachusetts B-LOC
02481 O
doing O
business O
as O
" O
Silicon B-LOC
Valley I-LOC
East I-LOC
" O
and O
AKAMAI B-ORG
TECHNOLOGIES I-ORG
, O
INC O
( O
" O
Borrower B-PER
") O
```

Output prompt

Figure 9: One-shot examples for data annotation prompting.

D Appendix: Compliance with Generative AI Policy

ChatGPT was used as a supplementary aid for understanding documentations, addressing basic Python code implementation and debugging questions, and producing Python code for data visualizations. In addition, Grammarly and ChatGPT were used to improve the clarity of the wording and correcting grammatical mistakes.

E Appendix: Project Code Access

Project code can be viewed by accessing GitHub repository: NLP-DataAdaptationProject